

Bulletin #31

POISONOUS SNAKES

Of Texas
And First Aid Treatment of Their Bites

By John E. Werler, *Director, Houston Zoological Gardens*



PUBLISHED BY TEXAS PARKS AND WILDLIFE DEPARTMENT, AUSTIN, TEXAS

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SNAKES OF TEXAS AND FIRST AID TREATMENT OF THEIR BITES

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POISONOUS SNAKES OF TEXAS AND FIRST AID TREATMENT OF THEIR BITES

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INTRODUCTION

This Bulletin has been prepared in response to many requests from hunters, ranchers, telephone construction personnel, Boy Scouts and others who spend much time outdoors, for a concise and illustrated guide to the poisonous snakes of Texas. The information presented is merely an outline of the poisonous snakes found within the state and is not intended to replace the several excellent snake books now on the market.

Sixteen species and subspecies of poisonous snakes, belonging to four general groups, are found in Texas. The most characteristic features of each are described. A photograph accompanies each description and a map indicating the snake's known range by counties is included. With this information, the reader should have little difficulty recognizing the poisonous snakes found in his region and, should he be required to render first aid for snake bite, the text and photos covering this subject will help him to do the job successfully.

A number of persons have extended help during the current revision of this booklet. I am most indebted to Alvin Flury, information and education officer for the Texas Parks and Wildlife Department, who not only did much of the necessary legwork for the project, but who also provided the impetus that saw it to completion. Thanks are due also to the following persons for county records of poisonous snakes in Texas: Ralph W. Axtell, Richard J. Baldauf, Edward W. Bonn, Bryce C. Brown, James R. Dixon, Alvin Flury, John W. Forsyth, W. C. Glazener, W. Grainger Hunt, L. M. Klauber, Verlin Nethery, Floyd Potter, Kirk Preston, Stephen Preston, Gerald G. Raun, Michael Sabath, Joe T. Stevens, Ernest C. Tanzer, Donald W. Tinkle, Clay Touchstone and John Wooters. Live specimens for making photographs used here were provided by Russell J. Long, Rusty Martin, Ernest C. Tanzer, Clay Touchstone and Lawrence Curtis. Finally, I wish to thank Robert L. Carlisle, Joseph F. Gennaro, Richard MacAllister, Don W. Micks, J. Fred Mullins, A. C. Stimson and John H. Werler for other information and assistance.

POISONOUS SNAKES OF TEXAS AND FIRST AID TREATMENT OF THEIR BITES

It is widely known among snake bite authorities that more persons die from snake bite in Texas than in any other state of the nation. This startling fact should encourage every Texan to obtain a thorough working knowledge of the correct first aid treatment for snake bite so he will be prepared to act promptly in the event of an emergency. In addition, and perhaps just as important, he should learn how to prevent a bite before it can happen. This means knowing where to expect poisonous snakes, how to avoid them, and how to identify on sight the venomous kinds found in the region where he lives.

PRECAUTIONS AT HOME

Statistics show that a large percentage of all bites takes place near the home and more than a few of these are inflicted upon small children playing in their yards. Therefore, let us consider first the possible presence of poisonous snakes around the home and some precautions that can be taken to keep them away.

Each year Texas zoos receive calls from distressed home owners within city limits who have discovered rattlesnakes or copperheads under their houses or beneath trash piles on their property. These people ask for assistance in the removal of the snakes or for advice which may prevent similar future invasions. The trespassing snake often can be disposed of quickly, but the removal of one snake does not always solve the problem. Something must be done to discourage other snakes from taking residence on the grounds.

The steps to be taken are simple; chief among them is the removal of all rubbish. Rock piles, trash piles, stacked lumber, tree stumps and other forms of debris near or under houses often harbor rats and mice which form the principal food of most snakes. In addition, this trash furnishes cover for the reptiles and offers them protection from enemies and bad weather. Therefore, if all rubbish is removed from the premises, the food and shelter which attract snakes are largely eliminated and the snakes are forced to seek a more suitable environment. Dense, low growing plants may also supply cover for snakes.

Finally, the removal of rubbish and dense vegetation enables the homeowner to easily see and destroy poisonous snakes that may be present. In settled areas, however, where poisonous snakes are particularly abundant and present a serious problem, it may be necessary to take further protective measures.

Dr. C. M. Bogert of the American Museum of Natural History has suggested the use of a quarter-inch mesh wire fence to keep snakes off residential

property. This yard-high, snake-proof fence is placed around the house in much the same manner as an ordinary picket fence, except that the bottom must be set about six inches into the ground to prevent snakes forcing their way beneath it. In addition, all gates must be provided with close-fitting sills on the bottoms and sides to insure a completely tight enclosure.

Experiments with fences of this kind were made to determine their effectiveness and to seek possible improvements in their construction, with the result that one important change was made. Copperheads and small rattlesnakes could not get over the vertically-straight fence, but a six-foot rattlesnake used in the experiment was able to climb over it. When the same fence was tilted outward at a 30-degree angle, not even the largest snake was able to reach the top. Although such fences are expensive and difficult to keep in good repair, they may be desirable under some circumstances.

A less costly method of keeping snakes away from residential property has been proposed by the manufacturers of a new chemical that allegedly repels and kills snakes. The product, a granular material with a civet musk odor, is reported by its distributors to kill a snake less than two feet in length if it is exposed longer than 20 seconds. Preliminary tests made at the Houston Zoo indicate that, in spite of claims to the contrary, the material had little effect on the snakes used. Each of more than a dozen different specimens, including examples of all the local poisonous kinds, unhesitatingly crawled through a wide barrier of the repellent that was poured on the ground. In some instances, the snake's tongue contacted the granules, but none of the test reptiles showed any ill effects from exposure to the chemical. While these crude tests are certainly not conclusive, it would seem advisable to use commercial repellents with some reservations until they can be proved effective. Meanwhile, we can still prevent most snake bites by observing a few simple safe practices.

PROTECTION IN THE FIELD

In the field, where poisonous snakes are more common, they present a greater hazard to human life; consequently, campers, crop farmers and others who spend a great deal of time outdoors should necessarily take more care in avoiding snake bites. Because almost all snake bites are inflicted on the arms or legs of the victim, these limbs require special protection. The use of a little caution, when placing hands or feet where snakes may be partially or completely hidden from view, is the best protection you can give them. This is particularly true when climbing hand-over-hand on rocky ledges, where your hands reach the level of the ledge before your eyes do. Rattlesnakes and copperheads are partial to such rocky hillsides and here, especially during the warm days of early spring, they prefer to coil and sun themselves.



Armadillo and pack rat burrows also make excellent shelters for rattlesnakes, and only a reckless person would find an excuse to reach into one of these holes. Yet one year, in South Texas alone, at least two snake bites occurred when the victims, each in search of small game, reached into armadillo holes and were bitten by rattlesnakes coiled there.

Another way to invite snake bite is to thoughtlessly turn over a log with bare hands or to step over one without first looking to see whether a snake is coiled on the other side. Many snakes, particularly the copperhead and coral snake, are fond of hiding beneath or within decaying logs, as any snake collector will testify, and such a log is at all times to be considered a potential snake den. If a log must be moved, use a long stick as a pry-bar. Stepping over a log will be less risky if boots or high-top shoes are worn, but even then it is safer to see first what is on the other side.

Several types of footwear offer good protection against the bites of most snakes. Especially effective are high-top leather shoes, riding boots, rubber boots or a combination of army “paratrooper shoes” and heavy leather puttees. Probably the best of these is a snakeproof boot made of extra 8 heavy bullhide leather, sold by the Gokey Company, 94 East 4th Street, St. Paul, Minnesota. For protection of the legs above the knees, snakeproof pants that weigh little more than ordinary duck trousers are available. They consist of three thicknesses of duck material and one layer of fine wire mesh, flexible enough to allow easy knee movement. Snakeproof leggings of similar material can be purchased for safeguarding just the lower legs. Recently marketed aluminum leggings furnish good protection in many cases, but some brands tested were too thin and easily damaged; others were too uncomfortable.

If a poisonous snake is discovered close by, the best protection is to remain as still as possible until the snake has moved away. It should be remembered that a snake is quick to strike at a moving object, so to quickly step away at such a moment may be disastrous. If a rattlesnake is heard nearby but cannot be located, do not begin a wild dash for safety. Location of the snake may be misjudged and by taking a step you are likely to walk into, rather than away from, it. Again, remain still until the snake is sighted and, when it is certain the snake is at least five or six feet away and no others are nearby, slowly back away. If you must move away, do so as slowly as possible.

Because our native poisonous snakes are mostly nocturnal in their activities, remaining hidden during the day and emerging at night in search of food, a flashlight should be used by persons who find it necessary to travel through snake country after dark. During the cool days of spring and autumn, however, nocturnal habits are often reversed. During the day snakes search

for warm spots in which to sun themselves; by nightfall they are again under cover.

None of our poisonous Texas snakes ordinarily can strike more than three-quarters of its body length, unless it has a firm backing or is striking downward from an incline. Certainly none has the ability to jump at an enemy, a feat often attributed to the rattlesnake. A snake on the defensive is coiled with the forward part of its body in a loose S position. When striking, this coil is straightened out and the head is thrust forward. It is not necessary for a snake to strike from a coil in order to bite. If picked up near the head, it may simply turn, open its mouth and bite the hand.

It is dangerous to believe, as many do, that a water moccasin cannot bite under water, and woe to the person who dares to seize a submerged cottonmouth.

RECOGNIZING THE POISONOUS KINDS

Suppose, in spite of attention to the precautions and protective measures just outlined, someone is bitten by an unidentified snake. Certainly the victim would not wish to delay treatment if the snake were poisonous, because every minute wasted would make recovery more difficult. On the other hand, he would not care to undergo either the pain associated with snake bite treatment or the anxiety following a bite if a harmless snake were the cause of the accident. Yet time and again people have been bitten by harmless snakes and have been needlessly given first aid and hospital treatment because the offending snake was incorrectly classified as poisonous. Still others have been bitten by harmless snakes and died from nothing more than fright. 9

When recognition of the snake is doubtful, it should, if possible, be killed and taken to a hospital or doctor's office for correct identification. Always keep in mind, however, that a recently killed poisonous snake may still be potentially dangerous. Even after its head is cut from the body, a reptile is capable of marked reflex activity; merely touching the head may cause it to bite. To be on the safe side, use a stick to lift or carry a dead snake.

It is evident that the ability to distinguish between harmless and poisonous snakes goes hand in hand with a knowledge of proper first aid treatment. Ability to recognize a poisonous snake on sight can best be achieved by a study of live poisonous kinds and a comparison of them with harmless species. At many zoos, where both kinds are on display in glass-fronted cages, they may be examined safely. If a zoo is not conveniently near, an examination of photographs showing the different kinds is probably the next best way to become familiar with them.

It is generally considered by the layman that four kinds of poisonous snakes are found in the United States—the rattlesnake, cottonmouth, copperhead and coral snake. This grouping, although somewhat arbitrary and certainly not zoologically correct, is firmly established in the minds of most people

and may very well be here to stay. Nevertheless, a more accurate and scientific approach is possible if we consider each species and subspecies as a different kind.

Illustrations in this booklet show the 16 kinds (species and subspecies) of poisonous snakes known in Texas. These photographs, together with descriptions in the text, should aid in identification. The accompanying distribution map for each form has been compiled from county records based on museum specimens, but in a few cases reliable "sight" records have been used as proof of a snake's occurrence within a county.

Distribution of snakes within Texas is incompletely known because of a lack of reliable records; for this reason, the distribution maps are necessarily far from complete. It is hoped, however, that the lists will be brought up to date as new localities come to light. If the reader can make any additions to the present lists, he is asked to communicate with the author or to deliver specimens to the nearest large college, zoo or museum maintaining a zoological collection. 10

Only 16 of the approximately 106 different kinds of snakes found in the state are dangerously poisonous to man, and some are so rare that they are seldom seen. In addition, we have in Texas several species of smaller snakes known as opisthoglyphs, each possessing a mild venom and a set of small grooved fangs far back in the upper jaw. Because of their weak and limited supply of venom and small fangs, which are poorly adapted for injecting poison into large animals, these rear-fanged snakes are considered harmless to man. By far, the majority of species are small, being not more than 15 inches long and about three-sixteenths of an inch thick. In this group are the blackheaded snakes (genus *Tantilla*) found over most of the state. The slightly larger spotted night snakes (genus *Hypsiglena*) of west and central Texas have enlarged but ungrooved teeth in the upper jaw. Two Mexican rear-fanged snakes, found as far north as the Brownsville region, are somewhat larger. They are the blackstriped snake (*Coniophanes imperialis imperialis*) which grows to about 20 inches in length, and the Texas cat-eye snake (*Leptodeira annulata septentrionalis*), which reaches a length of about three feet. Still another species, the very rare Texas lyre snake (*Trimorphodon wilkinsonii*), is known from extreme western Texas. It is generally less than three feet long.

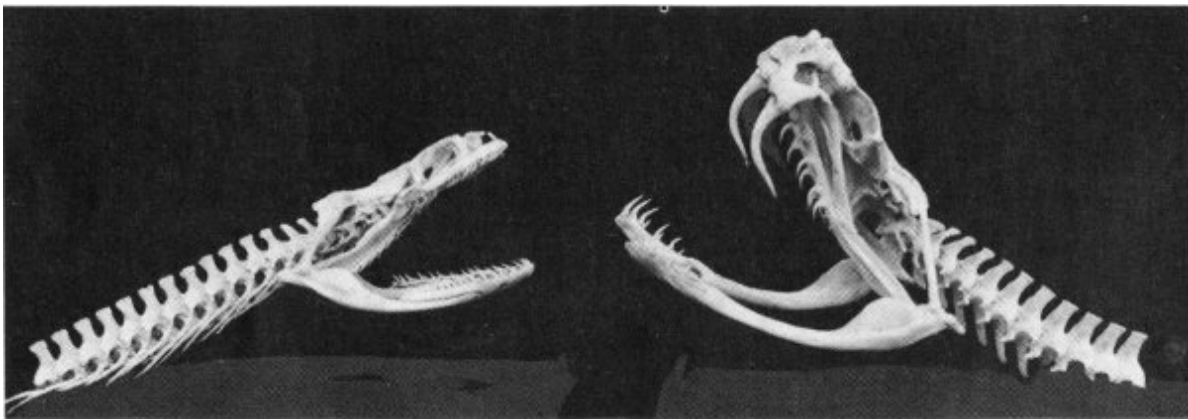
In spite of the small percentage of poisonous kinds of snakes in the state, it must be made clear that no one general rule can be used safely to identify all of them at a glance. It is a mistaken idea that all venomous snakes have broad, triangular heads. On the contrary, by using this rule, many of our harmless snakes look more dangerous than do some poisonous kinds. Furthermore, this generalization is made useless by several exceptions, a notable example being the coral snake which has a round head and does not look at all poisonous.

Even the characteristic rattle is not always present to make a rattlesnake's identification certain. Sometimes the snake's rattle is accidentally broken off, in which case this identifying appendage is gone. Then, too, a rattlesnake often is coiled in such a way that its rattles are concealed beneath a loop of its body. Again, the tell-tale rattle is not visible and it is necessary to recognize the snake by some other means.

Disregard all so-called "easy" rules by which poisonous snakes may be identified; instead, learn to know each one by its general overall appearance. For example, to identify the cottonmouth, look for a combination of its most characteristic features—relatively short, stout body and a broad, flat head. Also look for a body color of black, dark brown or olive and from 10 to 15 wide, usually indistinct, crossbands which are generally lighter in the center than on the edges. The upper jaw below the eye, as well as the lower jaw, will be light colored in contrast to the dark color above the eye. Together, these characteristics will make identification quite certain at a reasonable distance. Remember that the young of this snake are colored differently from the adults and you will not be able to identify the juvenile by using the color characteristics of the adult snake.

THE PIT VIPERS

Our poisonous Texas snakes belong to two families—the *Viperidae* (subfamily *Crotalinae*—pit vipers) and the *Elapidae* (cobra-like snakes). The pit vipers, which include rattlesnakes, copperheads and cottonmouths, possess an opening on either side of the head, between the eye and the nostril. With these heat-sensitive pits, the snake can locate warm-blooded prey in the dark and make a direct hit upon a rat or mouse which it cannot see.



Snakes belonging to this family are further characterized by the possession of elliptical eye pupils, somewhat triangular-shaped heads and long, movable fangs in the front of the upper jaw. Each fang is connected by a tube to the poison gland located just behind the eye. When the snake is at rest and its mouth is closed, the fangs lie folded back against the roof of the mouth. During the strike, the mouth is opened so that the upper and lower jaws form an angle of nearly 180 degrees. The fangs are then erected to point almost directly forward. With the fangs in this position, the strike results in a stabbing action rather than a true bite.

THE COPPERHEADS

Four kinds of copperheads are known in Texas. They are much smaller and more slender than the closely related cottonmouth and, because they have proportionately smaller fangs and less venom, are not very dangerous to man. The four copperheads resemble one another by having the same general pattern of chestnut or reddish-brown crossbands on a lighter body color. Differences are based mostly on the size and shape of crossbands and in the degree of marking present on the belly.

Copperheads are most common in rocky areas of hilly or mountainous country, as well as in wooded bottomlands. They are rarely seen in dry, cactus country. Although spending most of their time on the ground, copperheads occasionally climb bushes and low trees in search of food. Along the Colorado River near Wharton, they were observed high in trees; some were 40 feet above the ground. The snakes reached these unusual heights by crawling along wild grape vines that clung to the tree trunks and draped over lower limbs. One snake hunter in the area collected nearly 100 copperheads by pulling them off the branches.

Copperheads do most of their feeding at night. During the spring in some parts of the state, they are found in large numbers along streams and other moist areas, where they spend most of the day hidden beneath decaying logs and other debris which affords good cover. Copperheads also can be expected when air temperatures are high, and when the soil and vegetation are wet from recent rain. They often make their homes within the suburbs of large cities where, due to their nocturnal habits and protective coloration, they are apt to be overlooked.

Not quick to seek cover when approached, copperheads prefer to lie perfectly still until an intruder has passed. Once molested, they frequently vibrate their tails and, if among dry leaves, produce a buzzing sound not unlike that made by a rattlesnake. Copperheads are quick to strike at any annoying object, and often bite several times in rapid succession.

Copperheads seldom deliver a fatal bite because of their short fangs and small size. Their strike is often blocked by a mere trouser leg. Records of the Antivenin Institute of America show that during a 10-year period, not a single death resulted from 308 recorded copperhead bites regardless of the lack or kind of treatment given victims. In spite of these reassuring figures, it must be remembered that the copperhead is potentially a dangerous snake, especially when the bite involves a child. All bites from this species should receive the same urgent consideration given the bite of a diamondback rattlesnake.

The four forms of copperheads occurring in Texas are the only kinds found in the United States.

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SOUTHERN COPPERHEAD

Agkistrodon contortrix contortrix

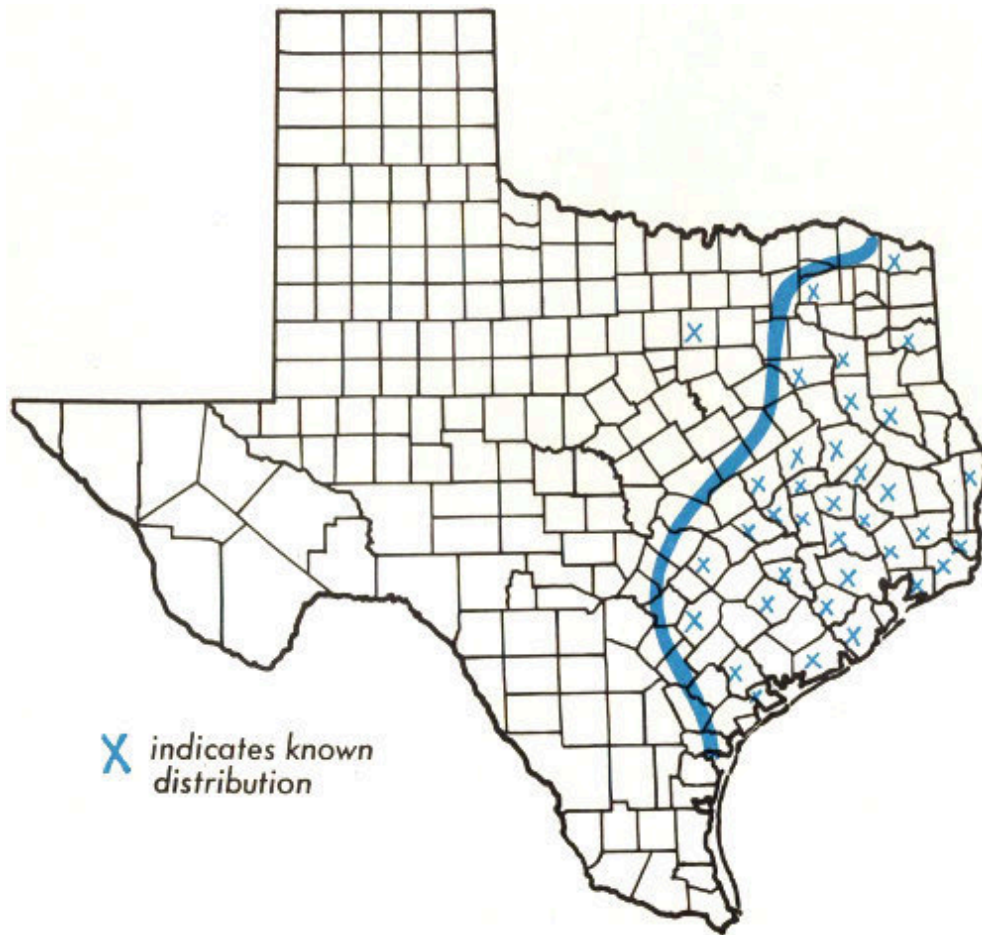


Description—Lighter in color than the other copperheads, this form has a body color of light brown or tan, often with a pinkish tinge. Hourglass-shaped crossbands of darker brown are very narrow along the middle of the back and are sometimes broken, forming two separate triangular markings, one on each side of the body. As on the northern copperhead, the bands are rounded at their bases. The belly is pale and indistinctly marked.

Size—Adults average between 20 and 30 inches long, while some especially large examples reach a length of more than 40 inches.

Young—The average number in a brood is five or six. As with all North American pit vipers, the young do not hatch from eggs but are born alive, enclosed in a thin membranous sack. They are paler than adults, with a more vivid pattern, and a bright sulphur yellow tail tip.

Distribution in Texas—The southern copperhead is known in about the eastern third of the state, where it has been found in the following counties: Austin, Bastrop, Bowie, Brazoria, Brazos, Burleson, Calhoun, Chambers, Cherokee, Colorado, Fort Bend, Gonzales, Grimes, Hardin, Harris, Harrison, Henderson, Hopkins, Houston, Jefferson, Leon, Liberty, Madison, Matagorda, Montgomery, Nacogdoches, Newton, Orange, Polk, Robertson, San Jacinto, Smith, Tarrant, Trinity, Victoria, and Walker.



NORTHERN COPPERHEAD

Agkistrodon contortrix mokasen

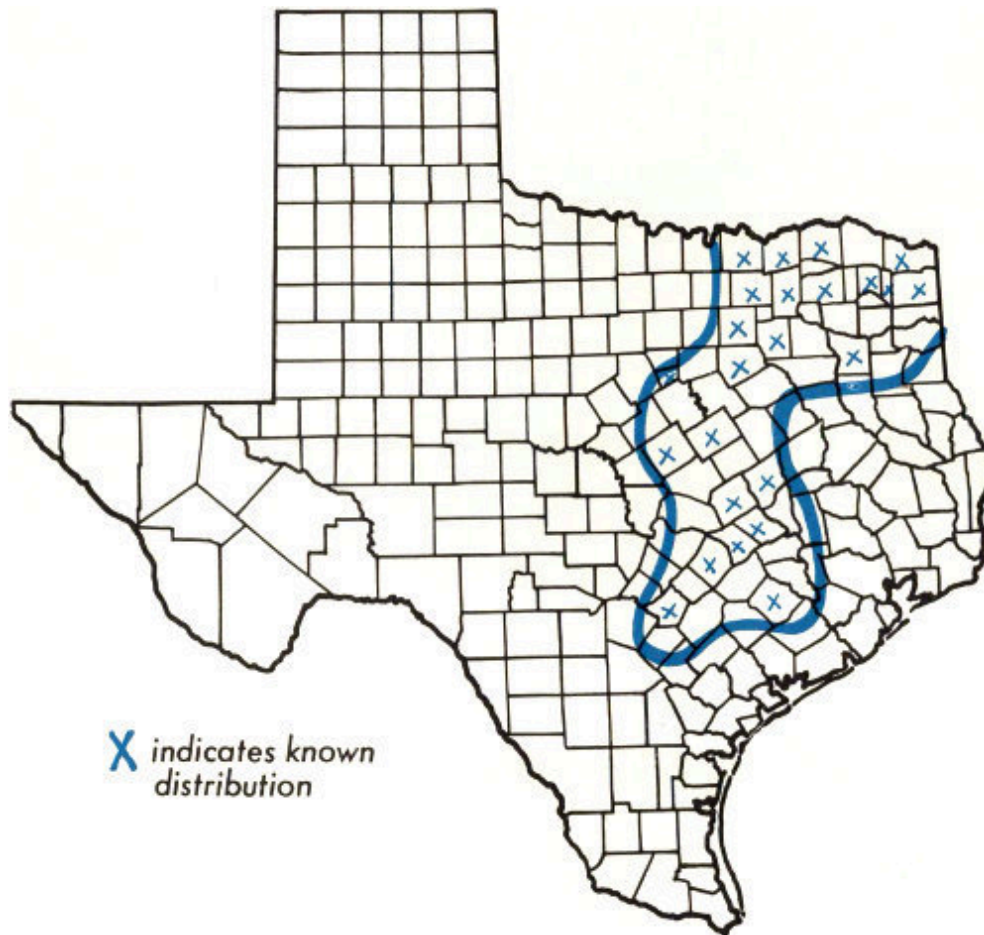


Description—This darker form has a pattern of reddish-brown or chestnut colored crossbands, which become narrower near the middle of the back and rounded at the bases. The darker color of these “dumbbell” shaped bands contrasts with the hazel-brown body color. The underside of the snake is dark and indistinctly mottled with gray or black. Its head is usually of a lighter tint than the body.

Size—Adults usually are about two and a half feet long but record size individuals of more than 40 inches have been caught.

Young—Newborn northern copperheads, from three to 14 in a brood, may be from eight to 10 inches long.

Distribution—Records of this snake are rather widely scattered over the northeastern part of the state. Specimens have been collected in the following counties: Bastrop, Bowie, Burleson, Cass, Collin, Colorado, Coryell, Dallas, Ellis, Fannin, Grayson, Guadalupe, Hopkins, Hunt, Kaufman, Lamar, Lee, McLennan, Milam, Morris, Robertson, Smith, Somervell, and Titus.



BROADBANDED COPPERHEAD

Agkistrodon contortrix laticinctus



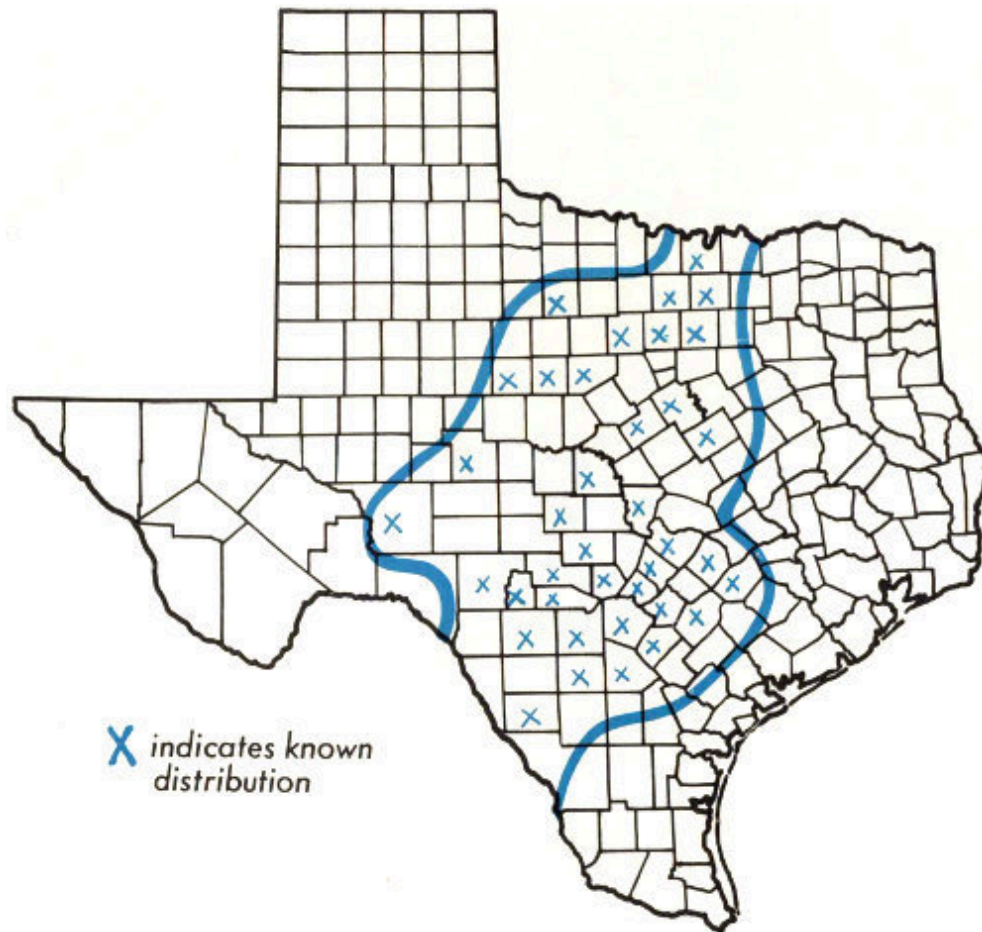
Description—The very broad, straight edged crossbands of this snake easily distinguish it from other copperheads. The similar Trans-Pecos copperhead differs by the presence of a light colored inverted U at the base of each crossband. The dark brown bands of this form are slightly narrower at the middle of the back than on the sides; their nearly straight edges gives the bands a squarish appearance. The dark crossbands contrast strongly with the lighter body color, while the belly is of almost the same shade as the bands. The tail is tipped with yellowish green, more pronounced in the young.

Size—Snakes of this subspecies probably reach a length of three feet but most adults are about two feet long.

Young—Litters probably average five or six but little else is known about the breeding habits of this form. 19

Distribution—Widely scattered records include the following counties: Atascosa, Bandera, Bastrop, Bexar, Bosque, Burnet, Callahan, Comal, Cooke, Crockett, Denton, Dimmit, Eastland, Fayette, Frio, Gillespie, Gonzales, Grayson, Guadalupe, Hamilton, Hays, Kendall, Kerr, Mason,

Medina, McLennan, Parker, Palo Pinto, Real, San Saba, Tarrant, Taylor, Throckmorton, Tom Green, Travis, Uvalde, Victoria, Wilson, Wise, and Young.



TRANS-PECOS COPPERHEAD

Agkistrodon contortrix pictigaster



Description—This species resembles the broadbanded copperhead by its straight edged, squarish crossbands but differs from all other copperheads by having uniformly dark or strongly mottled belly. It also differs from the broadbanded form by the presence of a light colored inverted U at the base of each crossband. The pattern consists of about 13 chestnut-brown crossbands with narrow, dark borders. The color between the bands is light hazel brown, flecked with darker brown.

Size—This is the smallest of the copperheads, probably not reaching a length of two and one-half feet.

Young—Nothing is known of the breeding habits of this rare snake.

Distribution—This form apparently is restricted to the mountains of West Texas, where it is known in Brewster, Jeff Davis, Presidio and Terrell Counties. Until 1949 it was found sparingly throughout its range and less than a dozen specimens had been found by experienced collectors. Recently, however, students from The University of Texas obtained about 100 of these snakes during a six weeks' summer course in a small section of

Terrell County. This indicates that the Trans-Pecos copperhead is common, but in restricted areas only. Such areas generally are in wooded canyons and live oak groves where there is some leaf litter.



THE COTTONMOUTH

Only one kind of cottonmouth is found in Texas. It is one of our heaviest and largest poisonous snakes and may reach a length of more than five feet. When viewed from above, the head appears triangular shaped and from the side the head looks flat on top.

The eye pupils, as with all members of the pit viper sub-family, are elliptical.

WESTERN COTTONMOUTH

Agkistrodon piscivorus leucostoma



Description—This is the most variable poisonous snake in the state, both in color and in the presence or absence of markings. Adults are much less brilliantly marked than are the young or newborn snakes. Adults have a background of dark brown and are marked by from 10 to 15 dark, wide crossbands somewhat lighter in the center than at the edges. The bands, which have irregular edges, become a little wider along the sides. 23 However, not all cottonmouths look like this. While some individuals have clearly defined crossbands, others have none. There is also considerable variation in color. Some cottonmouths are brown; some are olive brown or olive green; and some are entirely black. The lower jaw, as well as the upper jaw below the eye, is light in contrast to the dark color on top of the head. Young cottonmouths are vividly marked on a background of reddish-brown, highlighted by darker brown bands edged with white. They look much like copperheads.

Size—In Texas, this heavy-bodied snake reaches a maximum length of about four and a half or five feet, but the average is more nearly three feet.

Young—About eight are born in each litter. They are from six to eight inches long at birth.



Distribution—The cottonmouth has been reported in the following counties: Anderson, Aransas, Atascosa, Austin, Bandera, Bexar, Bowie, Brazos, Brazoria, Burleson, Burnet, Cass, Calhoun, Chambers, Cherokee, Collin, Colorado, Comal, Dallas, Denton, Ellis, Falls, Fannin, Fisher, Fort Bend, Galveston, Goliad, Gonzales, Grayson, Guadalupe, Hardin, Harris, Harrison, Hays, Henderson, Houston, Jasper, Jefferson, Kaufman, Kerr, Kimble, 24 Lamar, Leon, Liberty, Limestone, Marion, Mason, Matagorda, Maverick, McLennan, Medina, Montgomery, Morris, Nacogdoches, Newton, Nueces, Orange, Parker, Polk, Red River, Refugio, Robertson, San Jacinto, San Patricio, Shelby, Smith, Sterling, Tarrant, Tom Green, Travis, Uvalde, Val Verde, Victoria, Walker, Wharton, Wilson, and Wise.

The cottonmouth is one of our largest poisonous snakes. It is particularly abundant in the coastal marshes of southeastern Texas where it is found along streams, ponds and lakes. Although ordinarily rather sluggish, it

immediately draws back its head and opens its mouth widely in a threatening manner when annoyed, exposing the white tissue lining the inside of the mouth. This characteristic pose is responsible for its popular name. Moreover, like the copperhead, which also lacks rattles, it has the habit of vibrating its tail when sufficiently annoyed. Thus, when it is among dry leaves, or if the tail strikes a hard object, the resulting sound may be similar to that made by a rattlesnake.

The name water moccasin, which is loosely applied to any and all water snakes as well as to the cottonmouth, has resulted in a popular but misplaced belief that all “water” snakes are poisonous. It should be pointed out here that the cottonmouth, *Agkistrodon piscivorus leucostoma*, is our only poisonous aquatic serpent. The several kinds of harmless water snakes, which in some instances resemble it, are devoid of poison. They can, at most, inflict a bite no more serious than the scratch of a cat and these wounds require only the first aid recommended for minor cuts.

Harmless water snakes most frequently mistaken for the cottonmouth include the large diamondbacked water snake of central and eastern Texas, the blotched water snake found over most of the state, and the yellowbellied and broadbanded water snakes of eastern Texas. Most of them are relatively heavy bodied, possess somewhat diamond shaped heads and, although not venomous, will bite viciously if stepped upon or handled.

THE RATTLESNAKES

More than one-half of all the poisonous kinds of snakes known in Texas are rattlesnakes, and records show that almost every county has at least one variety. Texas rattlesnakes range in size from the very small, 18-inch western pigmy rattlesnake, which rarely if ever causes death among humans, to the seven-foot western diamondback rattlesnake, known to be one of the most dangerous snakes in North America. All have comparatively stout bodies, facial pits characteristic of the family, and rattles.

The rattle, which sets this snake apart, is a series of loosely interlocking horny segments which, when vibrated, produce a sharp buzzing sound as the segments strike against one another. A rattlesnake's age cannot be determined by the number of segments of its rattle because a new segment is added with each shedding of the skin. Because a snake may shed several times a year, the resulting number of segments added annually may be six or more. On the other hand, the segments are frequently broken off as the rattle becomes caught in underbrush, so that few rattlesnakes have a complete rattle.

WESTERN MASSASAUGA

Sistrurus catenatus tergeminus



Description—The general body color is brown or gray. About 39 dark brown, oval blotches extend along the middle of the back while two rows of smaller blotches are found along each side. These blotches are narrowly edged with a lighter color.

Size—Adults average two feet in length; larger specimens are sometimes as long as three feet.

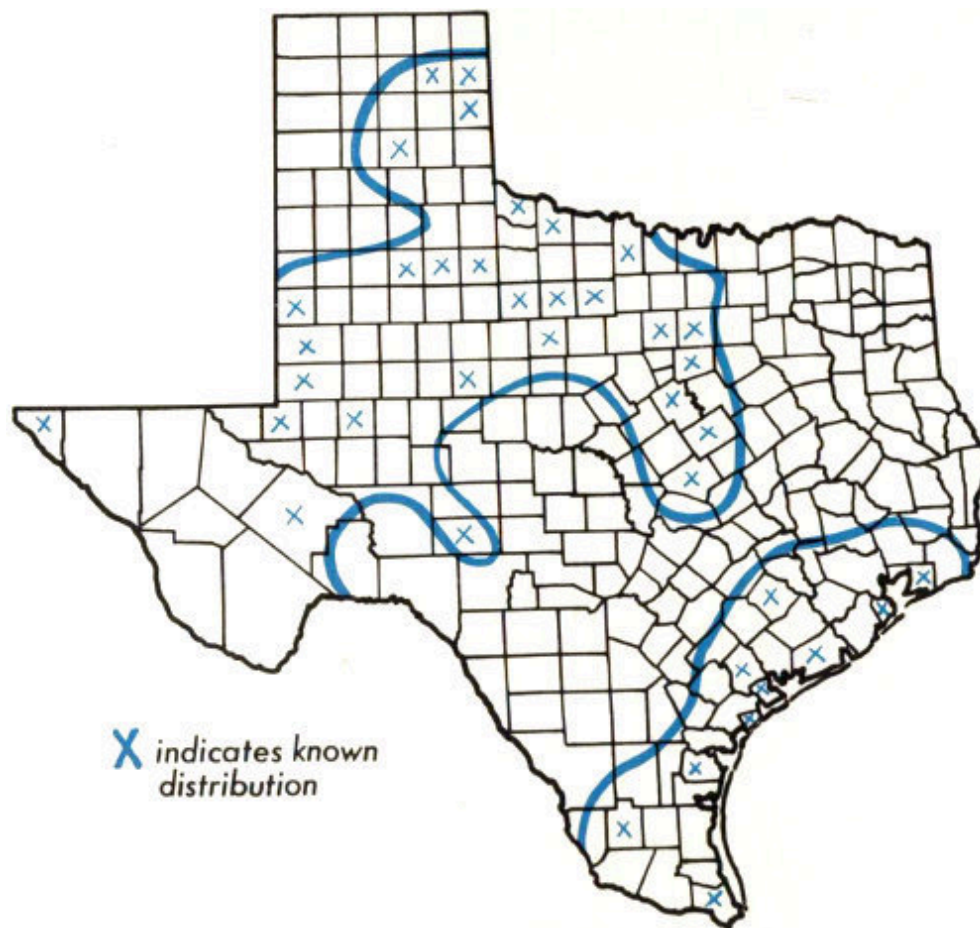
Young—The young resemble the adults but are lighter in color. Average number in a litter is eight or nine and they measure eight or nine inches at birth.

Distribution—Records of this snake are widely scattered throughout the state, including the following counties: Andrews, Aransas, Armstrong, Bell, Bosque, Brazos, Calhoun, Cameron, Chambers, Clay, Colorado, Crosby, Dickens, El Paso, Gaines, Galveston, Hardeman, Haskell, Hemphill, Jim Hogg, Johnson, King, Matagorda, McLennan, Midland, Nolan, Nueces, Parker, Pecos, Roberts, Shackelford, Sutton, Tarrant, Throckmorton, Victoria, Wheeler, Wilbarger, Winkler, Yoakum, and Young.

This small rattlesnake is uncommon in Texas, although years ago it was plentiful in some parts of the state. J. K. Strecker in his *Reptiles and Amphibians of Texas*, 1915, states:

Mr. Luttrell of Claude, Armstrong County, informed me that he has often killed from 50 to 60 during one wheat season, but during the past four or five years he has not seen more than half a dozen a year.

Wet places, usually near swamps or marshes, are its favorite habitat. One of our most docile rattlesnakes, it is seldom inclined to use its rattle, even when almost stepped upon.



WESTERN PIGMY RATTLESNAKE

Sistrurus miliarius streckeri



Description—The body color of this snake is from gray to grayish-brown. Its back is marked with about 35 small, dark spots which are wider than long. Another row of smaller spots is located along each side toward the belly.

Size—Specimens average 18 inches long and large examples may reach a length of more than two feet.

Young—The normal brood contains from eight to 10 but some may have as many as 18. Average length of the newborn is five or six inches.

Distribution—All county records of this snake, except one from 29 Mitchell County, are in the eastern part of the state. The Mitchell County record may be an error. Records include the following counties: Anderson, Angelina, Brazoria, Brazos, Chambers, Cooke, Dallas, Galveston,

Hardin, Harris, Harrison, Henderson, Houston, Jasper, Jefferson, Lamar, Leon, Liberty, Matagorda, McLennan, Mitchell, Montgomery, Newton, Orange, Red River, Refugio, Robertson, San Jacinto, Smith, Somervell, Victoria, Walker, Wharton, and Wise.

The western pigmy rattlesnake prefers to live in dry areas; reports of professional collectors indicate that it is found in greatest number after heavy rains and at night. While more aggressive than its close relative, the massasauga, its small size makes it one of our least dangerous poisonous snakes. Its rattle is small, often difficult to see, and cannot be heard at distances greater than a few feet.



WESTERN DIAMONDBACK RATTLESNAKE

Crotalus atrox



Description—Although individuals of this species show a great deal of variation in color, from a chalky white to a dull red, they can be identified immediately by the alternate black and white rings of about equal width on the tail. There is a pattern of brown diamond shaped markings along the middle of the back, which stands out against the lighter body color. Each diamond has a narrow light border.

Size—Adults are generally from three and a half to four and a half feet long. Reliable reports indicate that this species grows to a length of over seven and a half feet, and six-foot specimens are not rare.

Young—Broods average 10 or 12 but occasionally contain 20 or more. The newborn diamondback is about a foot long and looks very much like the adult.



Distribution—Known throughout most of the state except the extreme eastern part, it may be expected in many more counties from which records of its occurrence have not yet been received. It is definitely known in these counties: Andrews, Aransas, Archer, Armstrong, Atascosa, Bandera, Bastrop, Baylor, Bee, Bell, Bexar, Blanco, Borden, Bosque, Brazoria, Brazos, Brewster, Briscoe, Brooks, Brown, Burnet, Caldwell, Calhoun, Cameron, Clay, Coke, Comal, Comanche, Concho, Coryell, Cottle, Crockett, Crosby, Culberson, Dallam, Dallas, Dawson, Deaf Smith, DeWitt, Dickens, Dimmit, Donley, Duval, Eastland, Ector, Edwards, El Paso, Erath, Fisher, Foard, Frio, Galveston, Garza, Gillespie, Goliad, Gonzales, Guadalupe, Hardeman, Harris, Hartley, Hays, Hidalgo, Howard, Hudspeth, Hutchinson, Irion, Jack, Jackson, Jeff Davis, Jim Hogg, Jim Wells, Karnes, Kendall, Kenedy, Kent, Kerr, Kimble, King, Kinney, Kleberg, Knox, Lampasas, La Salle, Lavaca, Limestone, Live Oak, Llano, Lubbock, Lynn, Martin, Mason,

Matagorda, Maverick, McCulloch, McLennan, McMullen, Medina, Midland, Milam, Mills, Mitchell, Moore, Motley, Nolan, Nueces, Oldham, Palo Pinto, Pecos, Porter, Presidio, Randall, Real, Reeves, Refugio, San Patricio, Scurry, Shackelford, Somervell, Starr, Sterling, Stevens, Tarrant, Taylor, Terrell, Throckmorton, Tom Green, Travis, Uvalde, Val Verde, Victoria, Ward, Webb, Wells, Wilbarger, Willacy, Williamson, Wilson, Winkler, Wise, Wichita, Young, Zapata, and Zavala. 33

This is the most dangerous and, at the same time, the most common poisonous snake in the state. Therefore, it is one which all Texas outdoorsmen should be able to recognize on sight. A summary of snake bite cases in the United States over a 10-year period shows that more people died from bites of this species than from bites of any other North American snake. There are several reasons why this snake is responsible for so many bites and such a high number of deaths. Chief among these is its large size. It ranks as one of the two largest poisonous snakes in the country, being second only to the eight-foot eastern diamondback rattlesnake of the southeastern states. In direct proportion to its size, it has long fangs and poison glands which hold a great amount of venom. These factors insure a long strike and deep fang penetration. In addition, it has an unusually furious disposition and, if threatened with danger or sufficiently annoyed, will vigorously defend itself instead of seeking immediate escape.

Although strictly a land snake, the diamondback may sometimes be found crossing streams or ponds and, occasionally, individuals will venture out into lakes and bays.

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TIMBER RATTLESNAKE

Crotalus horridus horridus



Description—This snake is much like the larger canebrake rattler but lacks the dark stripe from the eye to the back of the mouth, and generally is without the reddish-brown stripe down the middle of its back. Dark brown chevron shaped crossbands contrast with the general body color of yellowish tan. In some specimens black stippling occurs between the markings. Both black and light color phases of this snake are found in some parts of its range. It is unknown whether the dark specimens occur in Texas. The tail is marked with three or four dark bands on the lighter specimens but is altogether black on the darker ones.

Size—Throughout its range this snake has an average length of from three and a half to four feet, but a specimen six feet long is on record.

Young—The number in a brood varies from three to 12. Length of the newborn is eight or nine inches.

Distribution—The timber rattlesnake prefers rocky hills and mountains that are not too heavily wooded, but it sometimes inhabits bogs and swamps at lower elevations. In either situation, it seldom survives for long in areas heavily populated by man. In common with most other rattlesnakes, it seeks escape when approached by man and fights only when surprised or cornered. Known in northeastern Texas, it is reported from Cooke, Denton, Eastland, Grayson, Lamar, Red River, Taylor, and Wise Counties.

Like the prairie rattler, it often seeks the same denning areas year after year and congregates in numbers to hibernate for the winter.



CANEBRAKE RATTLESNAKE

Crotalus horridus atricaudatus



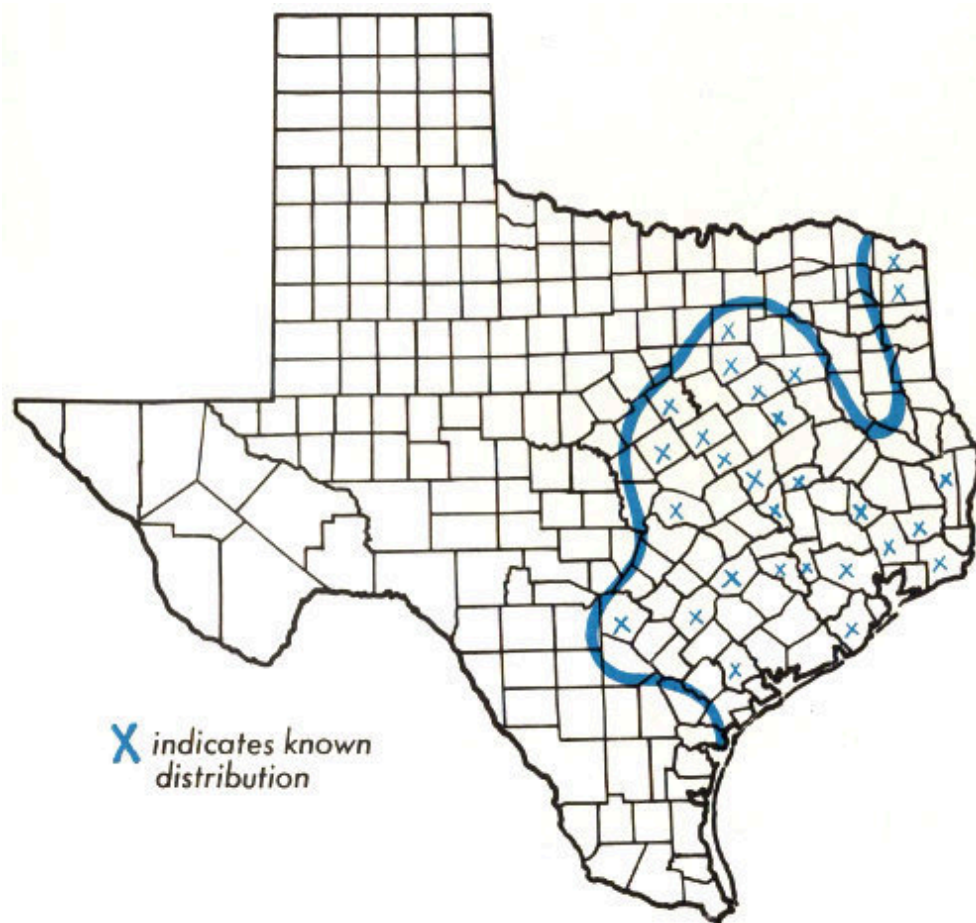
Description—This is a large and heavily built snake with a brown or tan back marked by a series of wide, dark, chevron shaped crossbands. It may be distinguished from the similar timber rattlesnake by its larger size, its more vivid markings, and the presence of a dark stripe from the eye to the angle of its mouth. Generally, a narrow, reddish-brown stripe extends down the middle of the back. Its tail, as in some other rattlesnakes, is entirely black.

Size—In Texas this snake reaches a length of about six feet, while in nearby Louisiana exceptionally large specimens, some nearly seven feet long, have been found. Length averages four and a half feet.

Young—About eight or ten are born in a brood.

Distribution—This species prefers wooded areas in wet bottomlands. It has been found in the following counties: Austin, Bexar, Bosque, Bowie, Brazoria, Brazos, Cass, Cooke, Coryell, Dallas, Denton, Eastland, Ellis, Falls, Fayette, Freestone, Gonzales, Grayson, Hardin, Harris, Henderson, Jasper, Jefferson, Liberty, Madison, McLennan, Navarro,

Robertson, San Jacinto, San Patricio, Taylor, Victoria, Waller, Williamson, and Wise.



BANDED ROCK RATTLESNAKE

Crotalus lepidus lepidus



Description—This snake is very similar to the mottled rock rattlesnake, with which it might be confused, but from which it differs by having a more mottled appearance between the crossbands and a dark stripe from the eye to the angle of the mouth. It has a pattern of about 18 or 20 widely spaced dark crossbands with irregular edges in contrast to its gray body color. Belly color varies from cream to pink.

In the Chisos Mountains, where there is much reddish igneous rock, the normal color of this snake is pinkish; the variety has been given the name of “pink rattler” by people of that region. Specimens from the limestone ledges along the Pecos Canyon at Howard Creek and Sheffield are very light in color, resembling the limestone rock on which they are found.

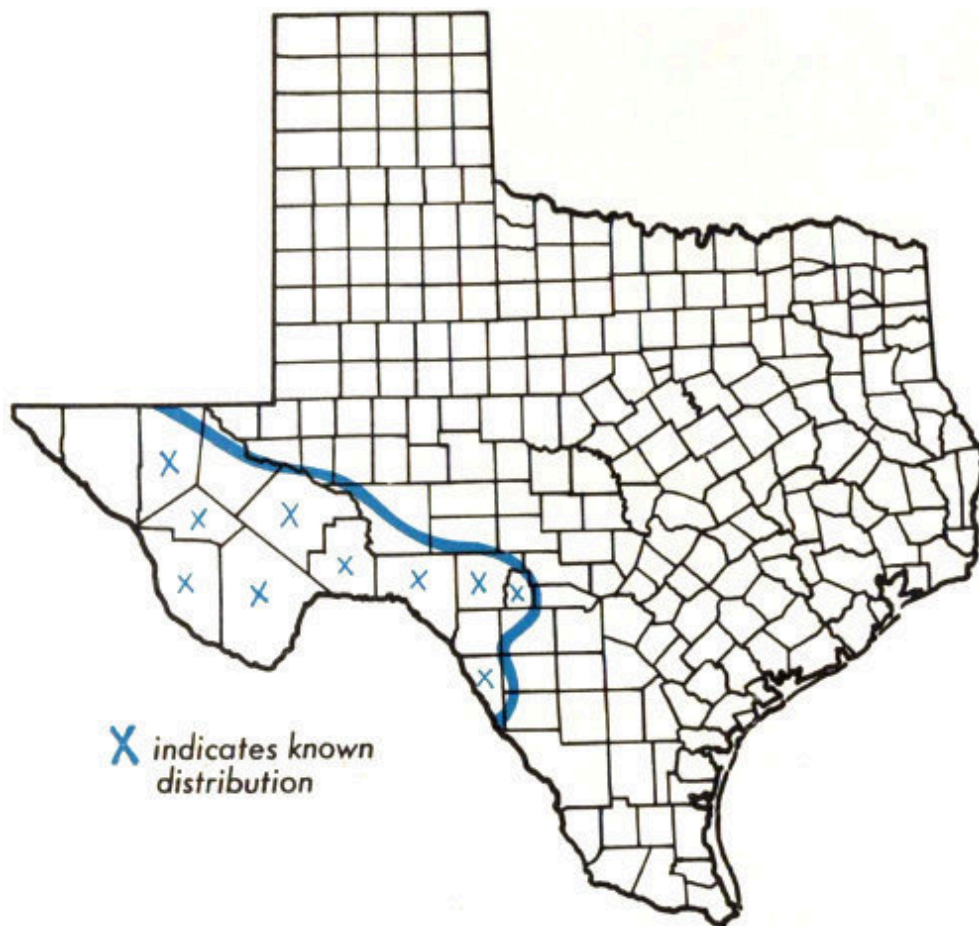
Size—This is one of our smallest rattlesnakes, having an average length of two feet. The rattle is rather large in proportion to the small size of the snake.

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Young—About four are born in a brood and they measure about seven and a half inches at birth.

Distribution—This form is restricted to the mountainous areas of the western and southwestern parts of the state; it has been found in the following counties: Brewster, Culberson, Edwards, Jeff Davis, Maverick, Pecos, Presidio, Real, Terrell, and Val Verde.

Because of its small size and distribution restricted to rocky places at high elevations, the banded rock rattlesnake cannot be considered a serious menace to man. Apparently there is no record of anyone ever having been bitten by this snake. It has a quiet disposition and, if alarmed, will immediately retreat within the masses of jumbled rock which are its home.



MOTTLED ROCK RATTLESNAKE

Crotalus lepidus klauberi



Description—This species and the banded rock rattlesnake look much alike. Both are small and slender, and marked with a series of widely-spaced dark crossbands along the length of the body and tail. The banded rock rattlesnake, however, has about 20 or 22 dark brown or black crossbands, which contrast strongly with its greenish-gray body color. It further differs from the banded rock rattlesnake in lacking a dark stripe from the eye to the angle of its mouth.

Size—Adults average two feet in length.

Young—A record of one brood is the only known published information concerning the young of this snake. Carl F. Kauffeld of the Staten Island Zoo mentioned a litter of four and wrote:

All were irritable from the first, promptly broke through the membranous sacs in which they were enclosed and struck violently at any passing object. All were marked and colored much like the adults except that the delicate pink along the venter of the latter was not in evidence; and the tails, which in the adults are salmon or terra cotta red (including the basal

segment of the rattle) were brilliantly sulphur yellow for at least their distal half....

The newborn snakes measured about eight inches in length.

Distribution—This subspecies is found in much the same type of country as the banded rock rattlesnake, being partial to rock slides high in the mountains. The two counties in the state in which it is known, El Paso and Culberson, are both in extreme western Texas. In disposition it is said to be quite timid; however, at times, it becomes irritable without much provocation.



BLACKTAILED RATTLESNAKE

Crotalus molossus molossus



Description—There are about 32 dark rhomboid markings along the back. These light edged blotches are centered with one or two irregular light areas, while the outer edges form bars which extend down each side to the belly. Body color varies from gray to olive green, while the blotches are dark brown or black. The tail is uniformly black, as its name implies.

Size—The blacktailed rattlesnake is one of our largest poisonous snakes. Texas specimens average three and a half feet but may grow somewhat longer. One, nearly 50 inches long, was collected at Persimmon Gap in Brewster County.

Young—About five young comprise the average brood.

Distribution—Records are available from the following counties: Bandera, Bexar, Brewster, Burnet, Comal, Culberson, Edwards, El

Paso, Hudspeth, Jeff Davis, Kendall, Kerr, Kimble, Medina, Pecos, Presidio, Real, San Saba, Terrell, Travis, Upton, and Val Verde.

Hilly areas with steep canyons are the preferred habitat of this snake. In the Big Bend region of Texas, where apparently it is the most common rattlesnake, specimens have been taken from as high as 7,400 feet elevation. It shows an inclination to coil in bushes or on tree limbs near the ground, although by far the majority of specimens encountered are found on rocky ledges. Its disposition has been reported by some to be very irritable, while others claim it is quiet and docile.



MOJAVE RATTLESNAKE

Crotalus scutulatus scutulatus



Description—This moderately slender snake is very similar in color and markings to the western diamondback rattlesnake. Its body is olive green with a pattern of darker diamond shaped markings down the middle of the back. These blotches are well defined by a border of light scales. It differs from the western diamondback in having narrower black tail rings and wider white spaces between these rings. A narrow light line extends from the eye to above the angle of the mouth.

Size—Average length is three feet, or shorter than the diamondback rattlesnake. Because of its more slender form, it does not appear as large as a western diamondback of equal length.

Young—The average litter contains eight, the young being similar to the adults in color and marking.

Distribution—The Mojave rattlesnake is known in Brewster, Hudspeth and Presidio Counties.

Unlike most other rattlers, this species is reported to be most active during daylight hours. It is rather quiet by nature and does well in captivity.



PRAIRIE RATTLESNAKE

Crotalus viridis viridis



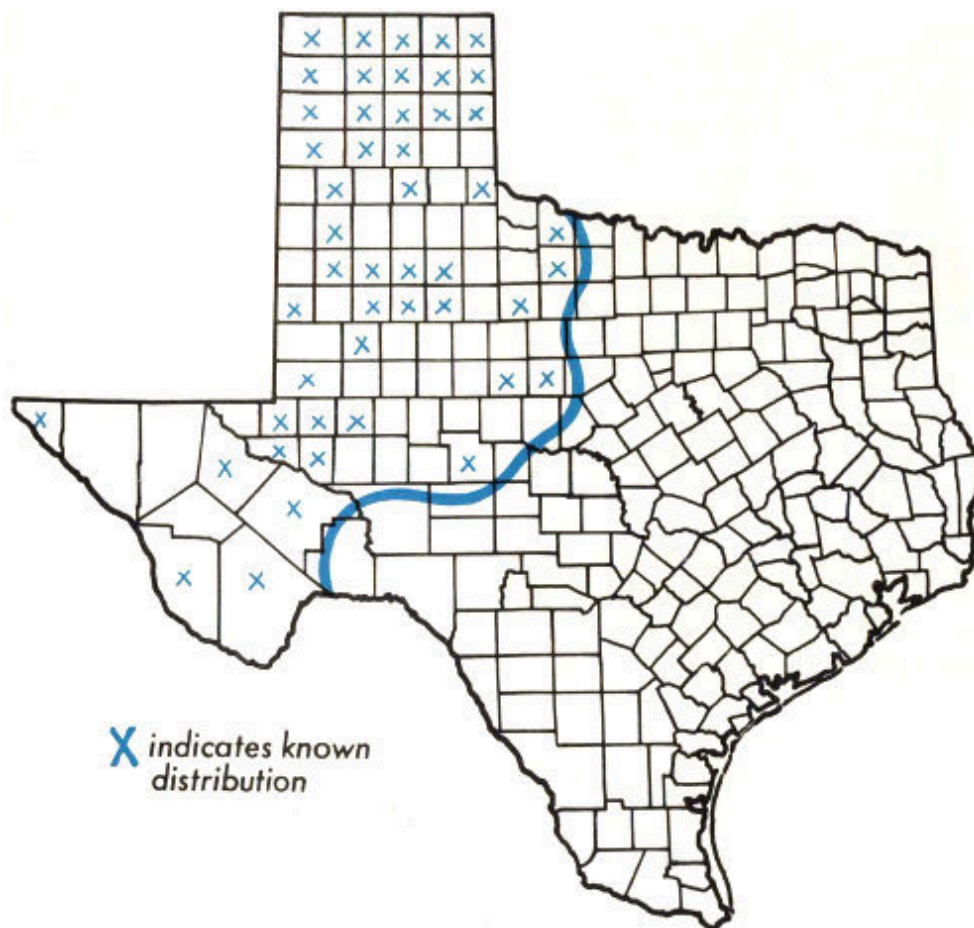
Description—Color of the back is greenish or grayish. A series of dark, rounded blotches extends down the middle of the back. These markings, with narrow white borders, become wider and shorter near the tail to form bands. In front of the eye is a narrow light line which extends backward and downward to the mouth.

Size—Rather slender in form, this snake reaches a maximum size of five and a half feet. Average length is three feet.

Young—Broods average 12 but vary from four to 21.

Distribution—Texas records of this snake are widely scattered. 47 Counties in which it is known include: Andrews, Armstrong, Baylor, Brewster, Briscoe, Callahan, Carson, Castro, Childress, Crane, Crosby, Dallam, Dawson, Deaf Smith, Dickens, Ector, El Paso, Garza, Gray, Hansford, Hartley, Haskell, Hemphill, Hockley, Hutchinson, Kent, Lamb, Lipscomb, Lubbock, Lynn, Midland, Moore, Ochiltree, Oldham, Pecos, Potter, Presidio, Randall, Reeves, Roberts, Sherman, Taylor, Tom Green, Ward, Wheeler, Wilbarger, Winkler, and Yoakum.

When aroused, this snake becomes a vicious adversary but, like most snakes, is satisfied to go its way if given half a chance.



THE ELAPID SNAKES

The single representative of this family (*Elapidae*) in Texas is small, slender, and brightly colored with rings of red, yellow and black. The head is small and rounded and eye pupils are circular. Its venom-conducting fangs, in the front of the upper jaw, are small and permanently erect; consequently, some chewing is required before the snake can inject its poison with certainty. As if to make up for its poor biting equipment, the coral snake possesses a venom of high toxicity, a venom much more potent than that of the pit vipers which are so well adapted for injecting their poison quickly and deeply. Because this poison produces scarcely any severe local symptoms, the danger from a bite may be overlooked and treatment delayed. Although pain is present, the usual dramatic symptoms of snake poisoning, such as extensive discoloration and great swelling at the site of bite, are scarcely noticeable. This absence of conclusive symptoms may lead the victim to believe he has been bitten by a harmless snake. In all coral snake bites, prompt action is necessary.

TEXAS CORAL SNAKE

Micrurus fulvius tenere



Left, coral snake; right, Mexican milk snake

Description—Generally less than two and a half feet long, this is our most colorful venomous snake and, at the same time, the least dangerous in appearance. Its small, narrow head, slender body and brightly colored pattern can be dangerously misleading. Children, especially, are inclined to pick it up because they are attracted by its colors and convinced that such beauty must be harmless. The pattern consists of red, yellow and black rings which encircle the body in the following order: a broad black ring, a much narrower yellow ring, a broad red ring, a narrow yellow ring, a broad black ring, and so on. Note that the red and yellow rings on the body touch one another. The snout is black and a broad yellow ring crosses the back of the head.

Identification of this snake would be simple were it not for the fact that several harmless snakes resemble it in form and coloration. These mimics are marked with yellow, red and black rings—but the arrangement is consistently different from that of the coral snake. The red and yellow rings of the coral snake touch one another, while in the harmless forms these

colors are separated by black rings. A simple rhyme adopted by Boy Scouts to help them associate “danger” with the color combination found on the coral snake is, “Red and yellow kill a fellow.” Remember this easy rhyme and, when you find yourself involved with a colorful little snake with adjacent red and yellow rings, be cautious.

The harmless kind most closely resembling the coral snake probably is the Mexican milk snake, technically known as *Lampropeltis doliata annulata* and found south of Kerrville. The head of this snake normally is black. Body and tail are marked with a series of from 19 to 25 narrow yellow rings bordered by slightly wider black rings. The red rings on the body are just as wide as the combined yellow and two adjacent black rings. Another of these mimics is the western milk snake (*Lampropeltis doliata gentilis*), a small species usually less than two feet long, which is native to central and western Texas. It has a pattern of from 25 to 40 yellow rings which are bordered by black. The red rings are separated from the yellow by black rings. All rings on this form are very narrow. In the southeastern part of the state is found still another of these mimics—the Louisiana milk snake (*Lampropeltis doliata amaura*). Rarely reaching a length of two feet, this form has a pattern of narrow yellow rings which are bordered on each side by narrow black rings. The much wider red rings are separated from the yellow by the black rings. In the scarlet snake (*Cemophora coccinea*) of eastern and coastal Texas, the pattern is a series of wide red blotches bordered by much narrower black bands. The yellow bands, about half the width of the red ones, are separated from them by the black bands. The belly is unmarked white or yellow. Adults are about a foot and a half long but a 25-inch specimen is on record.

Size—The coral snake is generally less than two and a half feet long with a body diameter of about three-eighths of an inch. The largest known specimen is nearly 42 inches long and was collected on the mid-Texas coast.

Young—This is the only poisonous Texas snake which lays eggs. From two to nine eggs constitute the egg complement of this form.

Distribution—This snake is known in the following counties: Angelina, Aransas, Atascosa, Austin, Bandera, Bastrop, Bee, Bell, Bexar, Bosque, Brazoria, Brazos, Brooks, Burleson, Burnet, Caldwell, Calhoun, Cameron,

Chambers, Colorado, Comal, Dallas, DeWitt, Duval, Ellis, Fort Bend, Galveston, Goliad, Gonzales, Guadalupe, Hardin, Harris, Harrison, Hays, Henderson, Hidalgo, Houston, Jackson, Jasper, Jefferson, Jim Hogg, Karnes, Kendall, Kenedy, Kerr, Kleberg, Lavaca, Lee, Leon, Liberty, Live Oak, Llano, Mason, Matagorda, McLennan, Milam, Montgomery, Morris, Nacogdoches, Newton, Nueces, Orange, Palo Pinto, Panola, Polk, Real, Refugio, Robertson, San Jacinto, San Patricio, Sutton, Tarrant, Terrell, Tom Green, Travis, Victoria, Walker, Waller, Wharton, Willacy, Williamson, and Wilson.

Unlike most other poisonous snakes, this species is a burrower, coming to the surface after a warm rain to feed upon small lizards and snakes. Contrary to even expert opinion, it is not primarily nocturnal, but becomes active during daylight and evening hours. It is often found in or under decaying logs or other trash, especially in damp regions; in the San Antonio area, specimens frequently are found under flagstones near homes.



In spite of its inoffensive nature, the coral snake is no different from other snakes in that it will bite if stepped upon or restrained. It does not deliberately coil and strike with accuracy like the pit vipers. Instead, it swings the forward part of the body from side to side until it can secure a hold to bite and then begins a chewing motion to imbed its short fangs. 51

The notion that its mouth is too small to bite effectively has in some instances resulted in careless disregard for its deadliness. It is true that the head appears very short and the mouth not capable of opening widely. Actually, the skull is rather elongated and the mouth can be opened to a greater degree than might be expected. Even when it bites a relatively flat surface, such as the back of the hand, the snake's closing mouth will pinch the skin, allowing the fangs to penetrate.

VENOMS

Generally speaking, snake venoms are divided into two broad categories, neurotoxic and hemorrhagic, depending on their destructive actions. Neurotoxic poison is characteristic of cobras and coral snakes and produces considerable pain but little or no swelling and discoloration at the bite. Death from this type of poison is the result of respiratory failure and is preceded by such symptoms as headache, muscular weakness, lethargy and facial paralysis with accompanying difficulty in speech.

Hemorrhagic venom, on the other hand, affects primarily the blood cells and vessels. Local reaction is evident soon after injection of the venom and consists of pain, discoloration, and swelling at the site of the bite. All of these symptoms gradually become more extensive. Weakness, nausea, vomiting and—occasionally—diarrhea may follow in a few hours. In many cases of snake bite, shock is present.

Although every snake's venom contains both the neurotoxic and hemorrhagic elements, the proportion of these components varies with each kind of poisonous snake. For example, the venom of the coral snake is primarily neurotoxic in action, but produces a small hemorrhagic effect as well. On the other hand, water moccasin venom, although basically hemorrhagic in action, has a greater amount of the neurotoxic element than do the poisons of the copperheads or rattlesnakes. Furthermore, neurotoxic effects are more evident following the bites of the Mojave and massasauga rattlers than they are in poisonings by other Texas rattlesnakes.



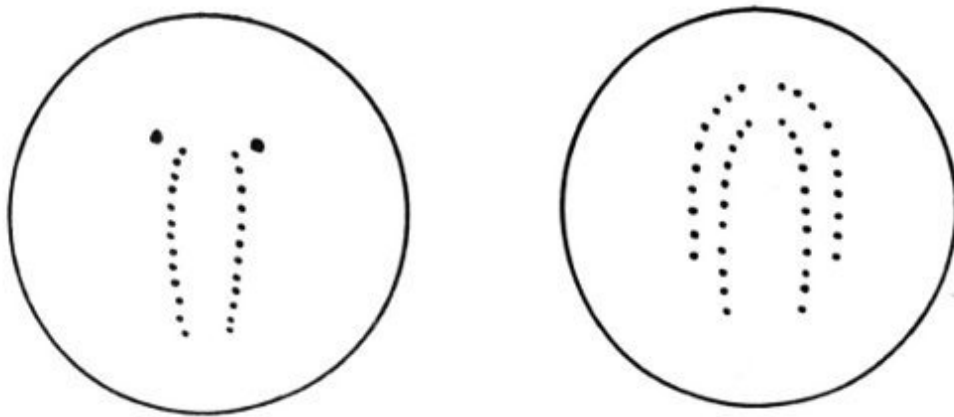
FIRST AID KIT

Equipment necessary for first aid treatment of snake bite includes a sharp cutting instrument such as a razor blade or sharp knife for making incisions, a constricting band to retard the flow of lymph, iodine or alcohol for sterilizing the knife and bitten area, and a suction device for removing the venom-contaminated lymph. This last item is especially important if you are alone and cannot reach the wound to apply suction by mouth. 53

Snake bite kits containing the essential items can be purchased at most drug or sporting goods stores at a cost of from two to five dollars, depending upon the make. Anyone who spends much time outdoors should carry a snake bite kit at all times, and he should know how to use it. A smaller kit can be more easily carried and is less likely to be left at home or in the car.

DIAGNOSING THE BITE

To make certain that a poisonous snake has caused the bite, first examine the wound for teeth marks. In a perfect bite by a poisonous snake (excluding the coral snake), the pattern will reveal two distinctly larger holes where the fangs have entered the flesh. There also may be two rows of smaller teeth marks between these punctures.



Furthermore, if the snake engages the lower jaw, two additional rows of small teeth impressions will be seen below the first group.

Frequently the bite pattern is not a clear one. For example, if the snake pulls to one side as it disengages the fangs after a strike, the result is a series of scratch marks instead of punctures. There is also the possibility that the snake will engage only a single fang; or, perhaps, during fang replacement, two fangs are temporarily in position on one or both sides of the upper jaw. Any of these conditions, as well as others, can contribute to an obscure bite pattern.

The bite of a coral snake often is difficult to diagnose by examination of the wound. Because this snake impels its fangs in a sort of chewing motion, the

pattern created by its bite may be two groups of closely spaced punctures where the fangs have entered the flesh a number of different times.

A clearly defined harmless snake bite pattern consists of a series of uniformly small punctures (four rows made by teeth in the upper jaw, two rows by teeth in the lower), but always without the large fang holes. More typically, a non-poisonous bite produces several rows of scratches.

Pit-viper poisoning is diagnosed primarily by the presence of *local* 54 signs and symptoms. The most important of these are:

1. *Pain* accompanies most poisonous snake bites. Generally intense and burning in character, it becomes more severe with the passing of time. This symptom alone is not conclusive because pain can be imagined following a non-poisonous bite. Occasionally in a severe bite, the pain is replaced by numbness and tingling.
2. *Swelling* at the bite area is present in every case of poisoning. It will appear within five to 30 minutes. In a severe case, the swelling may continue to spread for 24 hours. There will be no swelling from a bite by a non-poisonous snake or by a poisonous snake that injected no venom.



3. *Discoloration*, reddish or bruise-like in appearance, begins around the fang punctures within a half hour and becomes gradually more extensive.

Intense local pain is symptomatic of coral snake poisoning but, unlike a case of pit-viper poisoning, swelling and discoloration are not pronounced. Diagnosis of coral snake envenomation is difficult and must

be based primarily on systemic symptoms: headaches, weakness, lethargy and facial paralysis.

SEVERITY OF THE BITE

Even after the bite has been diagnosed as venomous, it is not immediately possible to predict the course it will follow. The severity of each case of snake poisoning is determined by the speed with which symptoms progress. Although the bite of a large snake is generally more serious than one caused by a small snake, the seriousness of a bite cannot be gauged by snake size alone. Some variable factors that affect the severity of each case include:

1. Age, size and health of the victim.
2. His allergy complex and sensitivity to protein poisoning.
3. His emotional condition immediately after having been bitten.
4. Location of bite on the victim.
5. Amount of fang penetration and venom injection.
6. Number of times the victim was bitten.
7. Kind and size of snake that bit him.
8. Whether or not the snake recently had eaten.
9. Conditions of the snake's fangs.
10. How soon treatment was administered.

FIRST AID TREATMENT

It is important that every snake bite victim receive first aid treatment as soon as possible. The patient must not exert himself by running, because increased circulation brought on by such physical activity will speed up absorption of the poison. For the same reason, the use of whiskey or other stimulants should be avoided.

The victim's state of mind is important; he must promptly be convinced that his chances for recovery are good—and, indeed, they are. He should not be terrified by the thought that every snake bite means certain death. Actually, a survey of case histories shows that with prompt and proper treatment, only about one or two per cent of all snake bites in this country are fatal.

There is a considerable difference of opinion about the correct first aid treatment for poisonous snake bite. The Division of Medical Sciences of the National Research Council, a section of the National Academy of Sciences, recently made a study to determine the most effective method of such treatment. It recommended immobilization of the bitten limb, application of a constricting band, and prompt incision and suction.

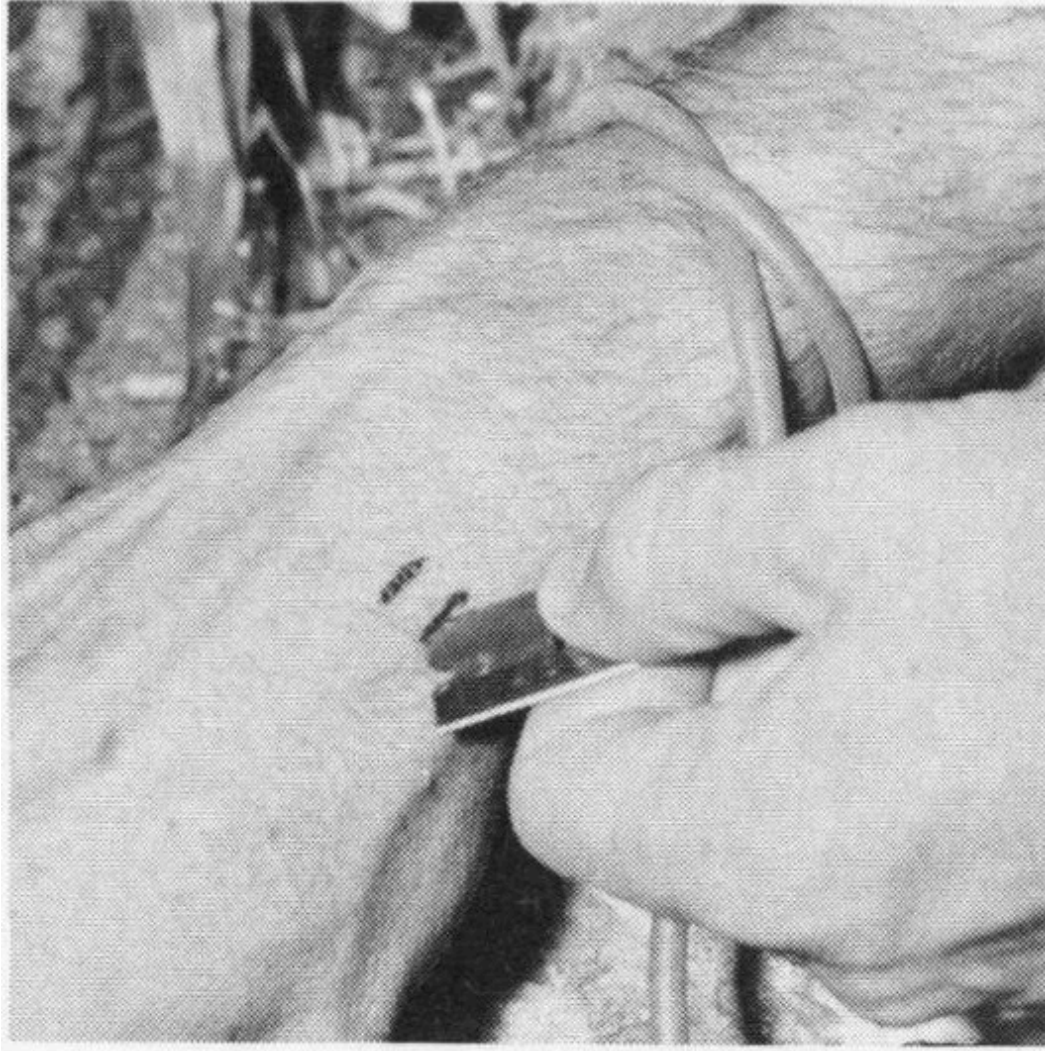
Based on these recommendations, the following first aid treatment for poisonous snake bite is suggested:

1. **IMMOBILIZE THE AFFECTED ARM OR LEG** whenever possible. Where this is not practicable, keep movement of the bitten limb to a minimum. Muscular activity helps increase the spread of venom. Whenever feasible, transport the victim by litter to further medical aid.
2. **APPLY A CONSTRICTING BAND** from two to four inches above the bite, between the wound and the heart. This will help to limit the spread of venom until it can be removed by incision and suction or neutralized by antivenin. A piece of rubber tubing or a strap tourniquet, included with every snake bite kit, is best for this purpose. When these are not available,

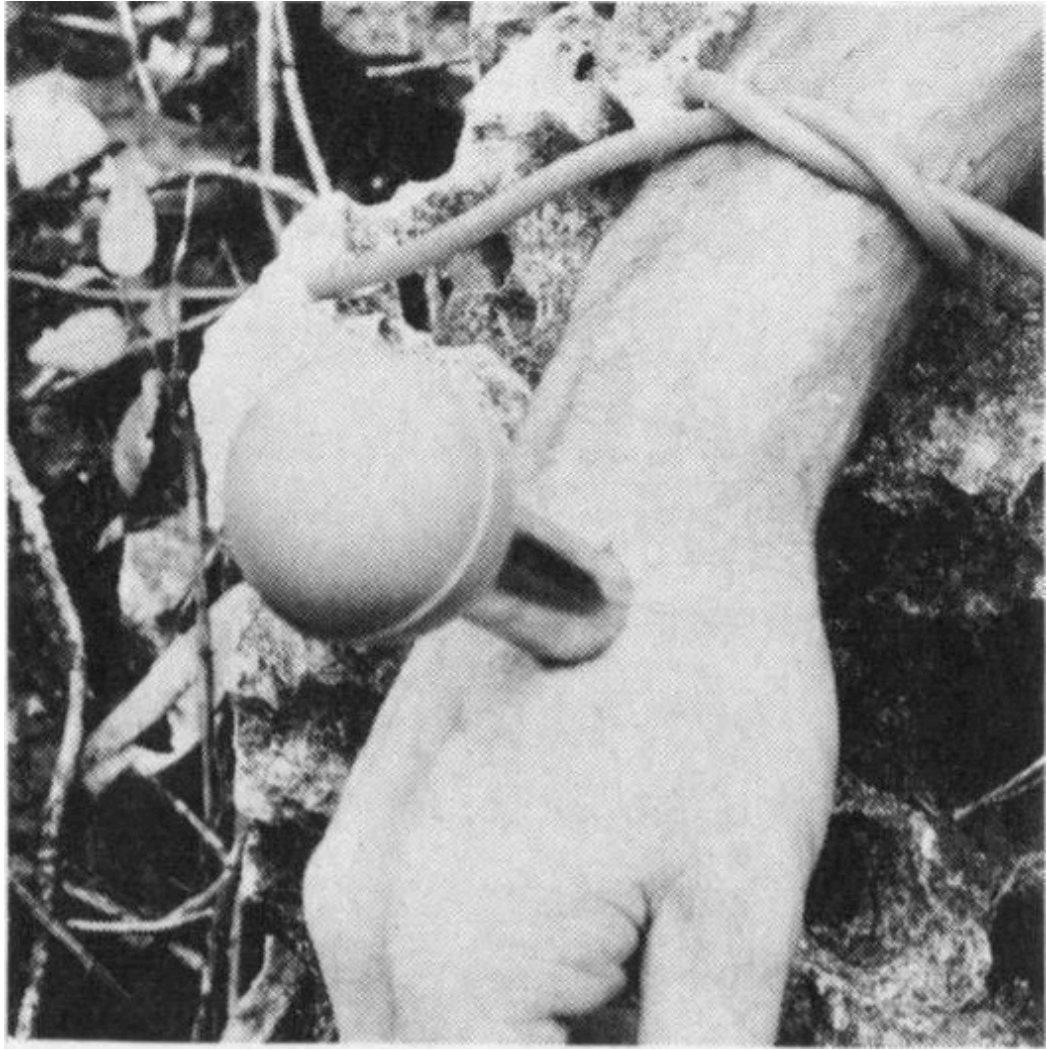
items of clothing may be used. A shoe lace, neckerchief or a strip of clothing torn from shirt or trousers will do. The poison, unless injected directly into a major blood vessel or deeply into a muscle, is absorbed slowly by the lymphatics below the skin. Therefore, do not restrict the deeper blood circulation by applying the constricting band too tightly. It should be loose enough for a finger to be slipped under it with little difficulty. Remember that during first aid treatment the constricting band must be loosened every 15 minutes for about two minutes. This precaution may prevent gangrene. If the bite is on the hand or forearm, take off rings, bracelets or other jewelry because subsequent swelling may make their removal difficult.

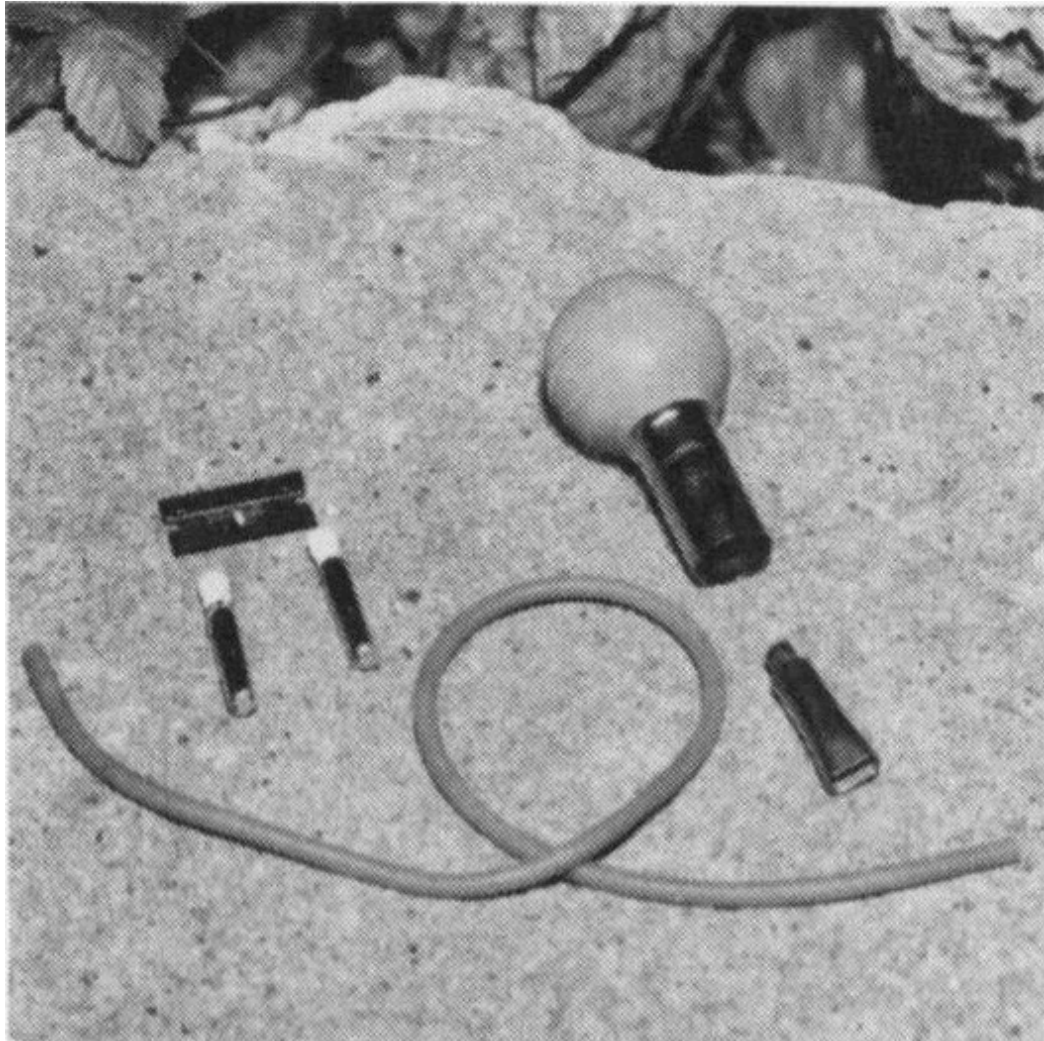


3. MAKE INCISIONS after sterilizing the cutting instrument and the bite area with iodine or alcohol. If no antiseptic is available, the blade of the cutting instrument can be sterilized by holding it over a flame (a match will do). Make *one* cut over each fang mark parallel with the long axis of the bitten limb, not across it. Incisions should be one-quarter inch long and one-eighth to one-quarter inch deep, but definitely no longer than the diameter of the suction apparatus being used. This would allow air to enter the suction bulb from the outside and the device would then be unable to work. Incisions are of the utmost importance to first aid treatment; without them little or no poison can be withdrawn from the wound by suction. However, making even a small incision involves some risk and this operation should be done with considerable care. Improper or carelessly applied first aid may actually do more harm than good. Although a physician may later decide to make additional incisions to relieve the pressure of swelling, only the cuts over the fang marks are recommended for first aid.



4. APPLY SUCTION to the cuts. This can best be done with one of the suction devices manufactured for that purpose but, if none is available, suction can be applied by mouth. There is little danger in oral suction unless the lips or inside of the mouth have cuts or abrasions. Contrary to popular opinion, a tooth cavity will not permit passage of venom 57 into the blood. Moreover, snake venom is destroyed by the stomach's digestive juices, so if some is accidentally swallowed, there is little need to worry if you have a healthy system. Only during the first 30 minutes following the bite can much venom be removed by incision and suction.





5. GET TO MEDICAL AID as soon as possible—but keep in mind that unnecessary physical exertion is harmful.

Antivenom may be administered soon after first aid has been started, but this is best left to a doctor. The North American Antisnakebite Serum made by Wyeth, Inc., of Philadelphia 3, Pennsylvania, is effective against pit viper bites but is of less value in the treatment of coral snake poisoning. Because coral snakes cause so few bites in the United States, no serum to neutralize their venom is prepared in this country. In South America, where these snakes are common and may reach a length of five feet, a serum to neutralize the poison is being produced by the Instituto Butantan at Sao Paulo, Brazil. It and other foreign snake bite serums often are available at larger zoos where exotic poisonous species are exhibited.

SUGGESTIONS TO PHYSICIANS

In most areas of Texas, snake bite is an uncommon medical emergency, and one with which few doctors have had experience. For this reason, the following recommendations by the National Research Council's Division of Medical Sciences are included for the benefit of the physician.

Statement on Hospital Care Following Bites by Venomous Snakes December 14, 1960

Admission Procedures

The routine admission history and physical examination should provide and record, if possible, the identity and length of the snake, the time of the bite, and the details of all first aid measures employed, including the time lapse for each and the mode of transportation to the hospital. The record should state whether a tourniquet, incision and suction, or the ligature-cryotherapy technique has or has not been used. Inquiry should be made concerning previous bites, allergic manifestations in general, and whether or not the patient had previously received horse serum. The admission examination should provide information from which the severity of the envenomation can be estimated as a guide to the need for the administration of antivenin and other therapy. Sensitivity tests should be instituted promptly during admission if not previously begun.

When the patient enters the hospital, blood should be drawn immediately for typing, matching and coagulation studies.

Although envenomation by one of the snakes of North America may present severe signs and symptoms, death is rare except in children or following envenomation by a large snake. However, permanent damage of an involved extremity is frequent following a bite by certain of the North American venomous snakes; plastic or orthopedic surgical repair to restore

function, or amputation, are not unusual consequences. Early and continuing close observation is needed to determine if certain therapeutic measures prevent or promote undesirable results.

Laboratory Tests

No rigid set of rules regarding therapy can be justified; the responsible physician must use his best judgment in his choice of tests to be performed as a guide to procedures to be used.

Clinical studies could include items such as repeated hematologic tests, hepatic and renal function studies, serial electrocardiograms, electroencephalograms, and other studies to therapy, depending on the composition of the venom involved and within the limits of personnel, time and equipment available.

Therapeutic Procedures

A. Systemic

1. Immobilization. During transportation, admission procedures, and most of the early hospitalization period, immobilization of the affected part and absolute rest should be continued. A sedative or analgesic may be administered to relieve restlessness and anxiety; ice bags may be applied to alleviate severe pain. The extremity should be immobilized in the position of function, and active and passive exercises to prevent contracture started after the third day if consistent with the patient's condition.

2. Blood Transfusion. Postmortem examinations have at times revealed extensive retroperitoneal and intraperitoneal hemorrhage, and hemorrhage into the viscera, including the liver and kidney. Progressive decrease in the total volume of circulating red blood cells has been attributed to the development of a massive hemolytic anemia or internal hemorrhage. Blood transfusions may be necessary and at times have been followed by marked improvement. Studies of the several factors involved in blood

clotting may be useful as guides to treatment. The first and subsequent specimens of urine should be especially examined for the presence of red blood cells, hemoglobin, and protein.

3. Electrolyte Balance. Abnormality of fluid and electrolyte balance should be detected and corrected on a continuing basis.

4. Antivenin. Polyvalent or specific antivenins prepared from venoms of snakes in the same geographic area should be administered in therapeutic quantity as recommended by the manufacturer only with full realization that the hazard of immediate allergic reaction or delayed serum sickness are factors to be evaluated in the decision to carry out this type of treatment. During hospitalization, antivenin should be given intravenously, provided that sensitivity tests indicate that the patient 59 is not allergic to the antiserum to be used. Desensitization, if necessary, should precede the administration of antivenin by any route. The use of antivenin in such cases should be carefully evaluated. Injection in normal muscles in other extremities would be the second choice, since local injections into the deposit site do not diffuse efficiently into the entire damaged area and would increase the hazard of pressure ischemia leading to increased tissue necrosis. Epinephrine should be available for immediate use when foreign protein is being administered.

5. Corticosteroids. The use of corticosteroids should be restricted to the prevention or treatment of late manifestations of allergy following administration of antivenins.

6. Antibiotics. A broad-spectrum antibiotic should be administered promptly in appropriate dosage if the reaction to envenomation is severe. Since the nature of the injury markedly predisposes to infection, and pathogenic bacteria are found in the wound, this use of antibiotics seems justified. Laboratory sensitivity tests, if available, should control the continuing choice of antibiotics to be used. A massive wound infection with severe systemic reaction could be mistaken for severe envenomation. Repeated blood and wound cultures would be of help in making the distinction.

7. Tetanus Prophylaxis. Tetanus toxoid should be administered upon admission if it has not been given as a first aid measure.

8. Respiratory Paralysis. If respiratory paralysis develops following envenomation by one of the *Elapidae* (this family includes the coral snake), the use of tracheostomy and intermittent positive pressure artificial respiration is indicated.

9. Renal Shutdown has been an occasional occurrence following massive envenomation. An awareness of this possible complication can do much toward the prevention and treatment of secondary effects arising after its occurrence. Routine daily tests such as B.U.N., CO₂ combining power, and serum potassium levels are indicated in severe cases.

B. Local Measures During Hospitalization

1. Tourniquet (Constricting band). A tourniquet applied following a bite by a venomous North American snake should be removed if envenomation seems mild or after a potent antivenin is given in therapeutic quantity. The prolonged use of a tourniquet or of a constricting band would increase local tissue damage due to the action of venom and might delay the vascular transport of antivenin into envenomated areas.

2. Incision and Suction. Substantial amounts of venom can be removed during the first half-hour from subcutaneous deposits by incision and suction. On the other hand, if the casualty is admitted to the hospital one hour or more following envenomation, an attempt to remove venom by incision and suction at the site of the bite would be of little value. However, if marked subcutaneous pitting edema develops, 60 interstitial pressure can be relieved by several longitudinal incisions extending into the subcutaneous tissues. Suction and fasciotomy may be required at times.

Parrish (1961), using a modification of Wood, Hoback and Green's (1955) clinical classification of pit viper venenations, has proposed the following guide for determining the severity of a poisonous snake bite. This classification is based upon present signs and symptoms and the clinical

course of the patient during the first 12 hours of hospitalization. It will be useful as a guide in treatment.

Grade O (No venenation). Fang or tooth marks present, minimal pain, less than one inch of surrounding edema and erythema, and no systemic involvement.

Grade I (Minimal venenation). Fang or tooth marks present, moderate pain, from one to five inches surrounding edema and erythema in the first 12 hours after bite, and no systemic involvement.

Grade II (Moderate venenation). Fang or tooth marks present, severe pain, six to 12 inches of surrounding edema and erythema in the first 12 hours after bite, with systemic involvement—nausea, vomiting, giddiness, shock or neurotoxic symptoms present.

Grade III (Severe venenation). Fang or tooth marks present, severe pain, more than 12 inches of surrounding edema and erythema in the first 12 hours after bite, with systemic involvement (as in Grade II).

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FILMS

A 16 mm. motion picture film in color, entitled, "Poisonous Snakes," is available from the Texas Parks and Wildlife Department, Reagan State Office Building, Austin, Texas. It deals with snake bite prevention, poisonous snake identification and first aid treatment.

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